

Introduction to Machine Learning

Lecture 3 Linear Regression - Statistical Viewpoints

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Outline

Review

- ▶ **Linear Algebra in linear regression:** Linear system, Residual minimization, projections in linear spaces
- ▶ **Implementation of linear regression:** Derivatives, gradients, and SGD algorithm
- ▶ **Polynomial regression:** Formulation, rationality, learning

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Today

- ▶ Statistical viewpoints of linear regression — Maximum Likelihood Estimation (MLE) or Maximizing a posterior (MAP)
- ▶ Bias v.s. variance (underfitting, overfitting, ...)
- ▶ The first (and the simplest) regularized linear regression model — Ridge regression

Recall the Example of Linear Regression

► **Data:**

- $\mathbf{x}_n \in \mathbb{R}^2$: the sale and the revenue of the n -th company
- $y_n \in \mathbb{R}$: the stock value of the n -th company

► **Task:** Predict a company's stock value based on its sale and revenue.

► **Model:** A linear regression model with Gaussian noise

$$\begin{aligned} \mathbf{x} \in \mathcal{X}, \text{ (A linear space with norm)} \\ y = \mathbf{x}^T \boldsymbol{\theta} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad \Rightarrow \quad y \sim \mathcal{N}(\mathbf{x}^T \boldsymbol{\theta}, \sigma^2). \end{aligned} \tag{1}$$

► **Loss:** the mean-square-error (MSE) $\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2$.

► **Learning Task:** Minimize the MSE of the regression model

$$\min_{\boldsymbol{\theta}} \sum_{n=1}^N |y_n - \mathbf{x}_n^T \boldsymbol{\theta}|_2^2 = \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2. \tag{2}$$

where $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times 2}$

The Key of Statistical Modeling

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A Frequentist Viewpoint: Maximum Likelihood Estimation (MLE)

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A Frequentist Viewpoint: Maximum Likelihood Estimation (MLE)

Given a set of samples $\mathbf{X} = [\mathbf{x}_i]$, we assume that the samples are sampled from a distribution $P(\mathbf{x}|\theta)$ (a model with parameter θ).

- **Principle:** Assume a **deterministic** model, learn the model via maximum likelihood estimation (**MLE**):

$$\begin{aligned}\hat{\theta}_{MLE} &= \arg \max_{\theta} \underbrace{\log P(\mathbf{X}|\theta)}_{\text{Loglike}(\theta;\mathbf{X})} \\ &= \arg \max_{\theta} \sum_{i=1}^n \log P(\mathbf{x}_i|\theta) \quad (\text{i.i.d. Assumption})\end{aligned}\tag{7}$$

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- Deterministic model $\rightarrow$ (pointwise) MLE $\rightarrow$ optimization (e.g., GD or SGD).

Key Statistical Analysis for Ordinary Linear Regression

- Given arbitrary N D -dimensional features $\mathbf{X} \in \mathbb{R}^{N \times D}$ and their labels $\mathbf{y} \in \mathbb{R}^N$, an ordinary linear regression is

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- ▶ Essentially, the learning task is maximizing $p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta})$ (MLE).
- ▶ $\mathbf{X}$ are random variables, and a linear regression is interested in the expected value of Y conditioned on X based on a linear predictor, i.e., $\mathbb{E}[Y|X]$.

Random Variables and Instances/Samples

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- ▶ Multiple instances $\mathbf{x}$'s often lead to a matrix $\mathbf{X}$, and similarly, we denote $\mathbf{X} \sim P_X$.
- ▶ $\mathbb{E}_{P_X}[X]$ and $\mathbb{V}_{P_X}[X]$ are expectation and variance of the r.v. X .

Mean, (Co)variance, and Their Unbiased Estimation

Suppose that we observed a set of i.i.d. samples $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$, each $\mathbf{x}_i \sim P$

► Mean

$$\mu = \mathbb{E}_P[X] \quad (10)$$

► Unbiased estimation of mean: the average of samples

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \quad (11)$$

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► Variance

$$\sigma^2 = \mathbb{V}_P[X] = \mathbb{E}_P[(X - \mu)^2] = \underbrace{\mathbb{E}_P[X^2] - \mathbb{E}_P^2[X]}_{\text{Derive it}} \quad (12)$$

► Unbiased estimation of variance:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \mu)^2 \quad \text{or} \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \hat{\mu})^2 \quad (13)$$

Properties of Mean and Variance

- ▶ $\mathbb{E}[aX] = a\mathbb{E}[X]$
- ▶ $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$
- ▶ $\mathbb{V}[X + a] = \mathbb{V}[X]$
- ▶ $\mathbb{V}[aX + bY] = a^2\mathbb{V}[X] + b^2\mathbb{V}[Y] + 2ab\text{Cov}(X, Y).$

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] \quad (14)$$

- ▶ Linear combinations

$$\mathbb{V}\left[\sum_{i=1}^K X_i\right] = \sum_{i,j=1}^K \text{Cov}(X_i, X_j). \quad (15)$$

The Bias of Estimation

- ▶ Given a statistical model with parameter θ .
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 - ▶ The expectation of $\hat{\theta}$ over all possible data X generated by the ground truth model θ .
- ▶ $\text{Bias}(\hat{\theta}, \theta) = 0 \Leftrightarrow$ The estimator $\hat{\theta}$ is unbiased.

Toy Example 1: Is average an unbiased estimation of mean?

Given i.i.d. random variables $\{X_n\}_{n=1}^N$, with expectation μ and variance σ^2 .

- ▶ Sample average $\hat{\mu} = \frac{1}{N} \sum_{n=1}^N X_n$.

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- ▶ Sample average $\hat{\mu} = \frac{1}{N} \sum_{n=1}^N X_n$.
- ▶ We have

$$\begin{aligned}\text{Bias}(\hat{\mu}, \mu) &= \mathbb{E}_{X|\mu}[\hat{\mu}] - \mu \\ &= \mathbb{E}_{X|\mu}\left[\frac{1}{N} \sum_{n=1}^N X_n\right] - \mu \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}_{X|\mu}[X_n] - \mu \\ &= \frac{N\mu}{N} - \mu = 0\end{aligned}\tag{17}$$

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► The unbiased estimation is $\hat{\sigma}^2 = \frac{1}{N-1} \sum_{n=1}^N (X_n - \hat{\mu})^2$.

Toy Example 2: What is the unbiased estimation of variance?

► Sample variance $\hat{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N (X_n - \hat{\mu})^2$.

$$\begin{aligned} \text{Bias}(\hat{\sigma}^2, \sigma^2) &= \mathbb{E}_{X|\sigma^2}[\hat{\sigma}^2] - \sigma^2 = \mathbb{E}_{X|\sigma^2}\left[\frac{1}{N} \sum_{n=1}^N (X_n - \hat{\mu})^2\right] - \sigma^2 \\ &= \mathbb{E}_{X|\sigma^2}\left[\frac{1}{N} \sum_{n=1}^N [(X_n - \mu) - (\hat{\mu} - \mu)]^2\right] - \sigma^2 \\ &= \mathbb{E}_{X|\sigma^2}\left[\frac{1}{N} \sum_{n=1}^N [(X_n - \mu)^2 - 2(X_n - \mu)(\hat{\mu} - \mu) + (\hat{\mu} - \mu)^2]\right] - \sigma^2 \\ &= \mathbb{E}_{X|\sigma^2}\left[\frac{1}{N} \sum_{n=1}^N (X_n - \mu)^2 - \frac{2}{N}(\hat{\mu} - \mu) \sum_{n=1}^N (X_n - \mu) + (\hat{\mu} - \mu)^2\right] - \sigma^2 \quad (18) \\ &= \mathbb{E}_{X|\sigma^2}\left[\frac{1}{N} \sum_{n=1}^N (X_n - \mu)^2 - 2(\hat{\mu} - \mu)^2 + (\hat{\mu} - \mu)^2\right] - \sigma^2 \\ &= \underbrace{\mathbb{E}_{X|\sigma^2}\left[\frac{1}{N} \sum_{n=1}^N (X_n - \mu)^2\right]}_{\sigma^2} - \underbrace{\mathbb{E}_{X|\sigma^2}[(\hat{\mu} - \mu)^2]}_{\frac{1}{N}\sigma^2} - \sigma^2. \end{aligned}$$

► The unbiased estimation is $\hat{\sigma}^2 = \frac{1}{N-1} \sum_{n=1}^N (X_n - \hat{\mu})^2$. (What if μ is known?)

The Variance of Estimation

- ▶ Given a statistical model with parameter θ .
- ▶ Given a set of observed data x 's and each $x \sim P_\theta(X) = P(X|\theta)$.
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- ▶ When $\hat{\theta}$ is the unbiased estimation

$$\mathbb{V}[\hat{\theta}] = \mathbb{E}_{X|\theta}[(\hat{\theta} - \theta)^2] \quad (20)$$

The Trade-off Between Bias and Variance

- ▶ Suppose that θ is the ground truth parameter of a linear model
- ▶ A set of data $\mathcal{D} = (\mathbf{X}, \mathbf{y})$ are observed and yield

$$y = \mathbf{x}^T \theta + \epsilon = f_{\theta}(\mathbf{x}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (21)$$

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In theory, the expected error on an unseen sample $\mathbf{x}$ can be decomposed as follows:

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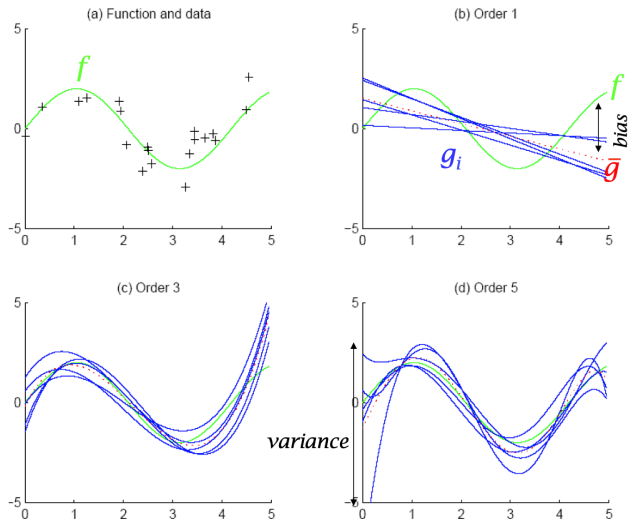
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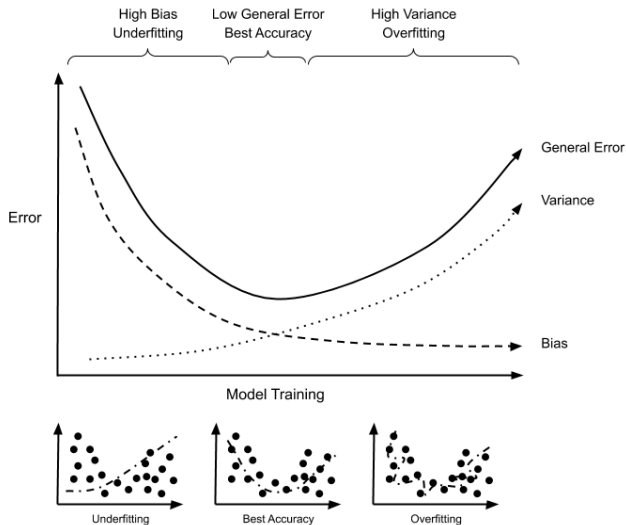
Bias-Variance Trade-off: An Indicator of Model Misspecification

Model misspecification:



Bias-Variance Trade-off: An Indicator of Training Process

Unsuitable stop criterion:



Overfitting and Underfitting

- ▶ **Overfitting:** Model complexity $\gg$ data complexity
 - ▶ The number of model parameters is larger than that of data points
 - ▶ **Case 1:** The model is wrongly complicated $\Rightarrow$ we need to simplify the model
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- ▶ How to learn complicated models from relatively sparse data?
 - ▶ Carefully-designed training process (SGD, early stop, dropout, ...), essentially, penalizing the variance/degradation of validation performance.
 - ▶ We need to impose side information on the model parameters (as **regularizers**).

Ridge Regression: MSE with L2 Regularization

- Ridge regression:

$$\min_{\boldsymbol{\theta}} \underbrace{\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2}_{L(\boldsymbol{\theta})} \quad (23)$$

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- Stochastic gradient descent:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \tau \nabla_{\boldsymbol{\theta}_t} L \quad (25)$$

Ridge Regression: A Bayesian Viewpoint

- Data model:

$$y = \mathbf{x}^T \boldsymbol{\theta} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (26)$$

- Model prior (Now the model itself is not deterministic any more, it is a r.v.):

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ML: Frequentist Statistic Viewpoint

Given a set of samples $\mathbf{X} = [\mathbf{x}_i]$, we assume that the samples are sampled from a distribution $P(\mathbf{x}|\theta)$ (a model with parameter θ).

- **Principle:** Assume a **deterministic** model, learn the model via maximum likelihood estimation (**MLE**):

$$\begin{aligned}\hat{\theta}_{MLE} &= \arg \max_{\theta} \underbrace{\log P(\mathbf{X}|\theta)}_{\text{Loglike}(\theta;\mathbf{X})} \\ &= \arg \max_{\theta} \sum_{i=1}^n \log P(\mathbf{x}_i|\theta) \quad (\text{i.i.d. Assumption})\end{aligned}\tag{29}$$

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- Deterministic model $\rightarrow$ (pointwise) MLE $\rightarrow$ optimization.

ML: Bayesian Statistic Viewpoint

- ▶ **Principle:** Assume a **probabilistic** model — the model θ yields a prior distribution.
- ▶ Bayes' theorem

$$\underbrace{P(\theta|\mathbf{X})}_{\text{Posterior}(\theta)} = \frac{P(\mathbf{X}|\theta)P(\theta)}{P(\mathbf{X})} \propto \underbrace{P(\mathbf{X}|\theta)}_{\text{Likelihood}(\theta)} \underbrace{P(\theta)}_{\text{Prior}(\theta)} . \quad (30)$$

- ▶ Maximum A Posterior (**MAP**) estimation

$$\hat{\theta}_{MAP} = \arg \max_{\theta} P(\theta|\mathbf{X}) = \arg \max_{\theta} P(\mathbf{X}|\theta)P(\theta) \quad (31)$$

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- ▶ Probabilistic model $\rightarrow$ (distribution) MAP $\rightarrow$ optimization (Variational Inference) or sampling (MCMC).
- ▶ The influence of prior decays with the increase of the number of samples.

Frequentist v.s. Bayesian

Frequentist

- ▶ **Pros:** more efficient in general, avoid the design of prior.
- ▶ **Cons:** non-robust to sparse data, not easy to quantify the uncertainty of the estimation (doable, but not efficient).

Bayesian

- ▶ **Pros:** prior makes it (relatively) robust to sparse data, quantify the uncertainty of the estimation (obtain the distribution of θ)
- ▶ **Cons:** require sophisticated design of prior, time-consuming in general.

In Summary

- ▶ A Frequentist viewpoint of linear regression
- ▶ Bias and variance in linear regression
- ▶ Over-fitting and under-fitting in statistical machine learning
- ▶ The first (and the simplest) regularized linear regression model: Ridge regression

Next...

- ▶ Regularization and A Bayesian viewpoint of regularized linear regression model
- ▶ Generalized Linear Model (GLM)
- ▶ Model/Feature selection strategies

Homework 1: DDL — March 16, 2025, Midnight

Python Programming (You can use AI for help)

1 Lab # 1 (3 Pts)

2 Lab # 2 (3 Pts)

Questions for Tech Report (6 Pts, **≤ 3 Pages**, You can use AI for help)

- Given N labeled samples $\{x_n \in \mathbb{R}, y_n \in \mathbb{R}\}_{i=1}^N$, we build a D -order polynomial regression model with data-dependent noise assumption, i.e.,

$$y = \sum_{d=1}^D x^{d-1} \theta_d + \epsilon, \text{ with } p(\epsilon|x) = \frac{1}{\sqrt{2\pi x^2}} \exp\left(-\frac{\epsilon^2}{2x^2}\right), \quad (32)$$

- 1 Learn the model θ via MLE, and derive its close-form solution. (4 Pts)
2 In addition, suppose that the elements of θ obey Laplace distributions, i.e.,

$$\theta_d \sim \text{Laplace}(0, b_d) \Leftrightarrow p(\theta_d) = \frac{1}{2b_d} \exp\left(-\frac{|\theta_d|}{b_d}\right), \quad \forall d = 1, \dots, D. \quad (33)$$

Learn the model θ via MAP, and derive its learning process. (4 Pts)

Delayed submission: -1 Pts per week.